The Interference Budget: fast weights, linear attention, and the synaptic memory of BDH

Concept. A *fast-weight* memory replaces the growing key–value cache of softmax attention with one fixed matrix `S`, updated by Hebbian outer products as tokens arrive: $S_i = S_{i-1} + k_i v_i^\top$, read as $out = S^\top q$. Seen as a running state, this *is* linear attention (Katharopoulos et al., 2020), and it is the mechanism Pathway's **Dragon Hatchling (BDH)** uses when it "reformulates attention as synaptic memory that updates as the model reads" (Kosowski et al., 2025).

Central claim. Because every binding is summed into the same matrix, reading back an earlier value returns the true value **plus a cross-talk term** $\Sigma_{j\neq i} (k_i \cdot k_j) v_j$. For random unit keys in `d` dimensions the expected recall error scales as **$\sqrt{(N/d)}$** for `N` stored bindings; it is near-zero only when keys are near-orthogonal, and average key correlation ρ adds a floor $\sqrt{((N-1)\rho^2)}$ that more dimensions cannot remove. A fixed state can therefore *process* an unbounded stream but cannot *retain* unbounded content: forgetting through interference is forced by linear algebra, not a tunable defect.

Design pressure and trade-off. Softmax attention keeps every past token in a KV cache, so memory and per-token compute grow with sequence length. Fast weights trade that cache for constant memory and linear-time inference. The cost is expressivity: a rank-`min(d,m)` state cannot reproduce arbitrary long-range retrieval the way a full cache can, which shows up on precise long-copy benchmarks (Yang et al., 2024).

Approach	State	Cost/token	Exact recall	Adaptation
Softmax + KV cache	grows O(t)	O(t)	yes, to context limit	in-context, lossless
Linear attention / fast weights (2020)	fixed d×m	O(1)	lossy, √(N/d) error	additive in state
Gated LA / DeltaNet (2024)	fixed d×m	O(1)	better: gating + error-correcting write	gated in state
BDH-GPU (2025)	synaptic σ over ReLU-low-rank features	O(1)	lossy, aided by ~5% sparse activations	Hebbian in synaptic state

A decay $\lambda<1$ buys recency by genuinely losing old information; the delta rule subtracts the current prediction before writing, cutting interference for repeated keys. RetNet, Mamba's state updates, GLA and Gated DeltaNet differ mainly along these two axes.

Where BDH and BDH-CQ sit. The Dragon Hatchling paper (arXiv:2509.26507, §6.1) defines BDH-GPU as "a specific kind of ReLU-lowrank feed-forward network, and a linear attention mechanism"; its positive activations are sparse at "about 5% level" (§6.4), and it matches GPT-2 on language and translation at equal parameters from 10M to 1B (§4.2). Pathway separately reports monosemantic synapses, a scale-free heavy-tailed neuron graph, 97.4% on Sudoku Extreme with no chain of thought (research post, Mar 2026), and pretraining scaling laws holding to ~600B parameters (largest *benchmarked* model in the family is 150M). **BDH-CQ** (arXiv:2608.09888) adds demonstration-driven latent reasoning: its state update $S_t = U\theta(S_{t-1}, D_t)$ is described as "generally related to attention, fast-weight memory, and linear-attention views of contextual association ... capturing the special case $S_t = S_{t-1} + U\theta(D_t)$" (§3.2) — exactly $S = \Sigma\ k_i v_i^\top$ with `i` indexing demonstrations. "Neither task identifiers nor evaluation-task demonstration pairs participate in training, and no parameters are updated at inference time"; a 150M model reaches 29.5% pass@2 on public ARC-AGI-1 at ~$0.0007/task. Contrast HRM/TRM, whose ARC pipeline is transductive and needs a per-task backward pass (§8). Its low/medium/high "effort" levels (pass@2 21.0 / 27.0 / 29.5%) spend more *recurrent* compute, not more chain-of-thought tokens.

Evidence, labelled. The linear-attention and associative-memory results — the √(N/d) capacity law, the cross-talk decomposition, the delta rule — are external and widely replicated; classical Hopfield capacity is ≈ 0.14·d (Hopfield, 1982), raised by a nonlinear read (Ramsauer et al., 2021). The ReLU-lowrank + linear attention formulation and the 10M–1B GPT-2 parity are *formal / in the paper*. Everything else specific to BDH — the ~5% sparsity measurement, monosemanticity, Sudoku 97.4%, ARC-AGI 29.5%, the 600B scaling figure, SageMaker HyperPod development — is *developer-reported* by Pathway or AWS, with no independent reproduction. A benchmark is not a deployment; the interactive toy accompanying this summary is a hand-built memory, not a BDH checkpoint.

Most important limitation. The √(N/d) law assumes the model has *learned* to place near-orthogonal keys; real linear-attention models often do not, so they forget more than the best case. And none of this is durable learning — the state lives for one session, and consolidating useful fast state into slow weights is an open problem the BDH authors name explicitly. Where BDH-CQ has no direct role — the capacity law itself — this summary says so rather than inventing a connection.

Read next: Katharopoulos et al., *Transformers are RNNs* (ICML 2020); Yang et al., *Gated Linear Attention* (ICML 2024) and *DeltaNet* (NeurIPS 2024); Kosowski et al., *The Dragon Hatchling* (arXiv:2509.26507, 2025) and *BDH-CQ* (arXiv:2608.09888, 2026); Ramsauer et al., *Hopfield Networks is All You Need* (ICLR 2021).